

# FINAL EXPLANATION: PHYSICS GRADE 10

## *Student Assessment Document*

### FINAL EXPLANATION: PHYSICS GRADE 10: GREENHOUSE EFFECT AND CLIMATE CHANGE

#### Student Assessment Document

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| Student Name | _____ |
| Class        | _____ |
| Date         | _____ |

### INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the unit. Use everything you have learned in Lessons 1 to 7 to answer the Driving Question: Why are Mount Kenya's glaciers melting and Kenyan farmers' rains becoming unpredictable, and what can the physics of our atmosphere tell us about how to respond? Answer each section in your own words in the space under the prompt. Use evidence from the unit, such as the Lewis Glacier images and data, the testimony of James Muriithi from Embu County, the PhET simulation and the Keeling Curve. Where it helps, draw a labelled diagram. Write in full sentences and use the correct physics words: shortwave radiation, longwave infrared radiation, absorb, re-emit, back-radiation, greenhouse gas, natural greenhouse effect and enhanced greenhouse effect. Link each answer back to Mount Kenya and the long rains.

### Part 1: What the Evidence Shows

Look back at the anchoring phenomenon from Lesson 1 and the data you met later in the unit. (a) Describe two pieces of evidence, with numbers, showing that the Lewis Glacier on Mount Kenya has changed. (b) Describe what James Muriithi, the maize farmer in Embu County, reported about the long rains and his harvests. (c) Explain why it is puzzling that a glacier on a mountain and the rains on a farm hundreds of kilometres away are changing at the same time, and what you would need to know to decide whether one cause could be responsible for both.

*Write your answer here:*

## Part 2: The Physics of the Greenhouse Effect

**Explain how the greenhouse effect works. In your answer, trace the path of energy from the Sun to Earth's surface and then back out toward space. Name the type of radiation at each step. Explain what greenhouse gas molecules such as  $\text{CO}_2$ , water vapour and  $\text{CH}_4$  do, and what you saw in the PhET simulation about  $\text{N}_2$  molecules compared with  $\text{CO}_2$  and  $\text{CH}_4$ . Then explain the difference between the natural greenhouse effect and the enhanced greenhouse effect.**

*Write your answer here:*

### Part 3: From Kenyan Activities to Climate Change

**Build a complete cause-and-effect chain that starts with a human activity in Kenya and ends with the retreat of the Lewis Glacier. Name at least two Kenyan activities that increase greenhouse gases and the gas each one releases. Use the Keeling Curve and the global temperature anomaly graph to say what evidence shows that rising greenhouse gases and rising temperature are linked. Say which of these links is shown by correlation in data and which is explained by physics.**

*Write your answer here:*

### Part 4: Why the Rains Are Becoming Unpredictable

**Use the physics you have learned to explain why the long rains in Embu County have become unpredictable. In your answer, (a) say what happens to evaporation when surface temperatures rise, (b) explain how this affects the water cycle and the rains over Kenya, and (c) explain why one cause can explain both the shrinking glacier and the farmer's experience. Finally, explain why the ozone layer is not the main cause of either change.**

*Write your answer here:*

### Part 5: Responding to the Problem

**The driving question also asks what the physics can tell us about how to respond. (a) Using your causal chain, suggest two Kenyan actions and explain at which step of the chain each one acts. (b) Explain what the Montreal Protocol shows about how the world can respond to atmospheric damage. (c) Name one question about climate change that you still have after this unit.**

*Write your answer here:*

| RUBRIC                                               |                                                                                                                                                                                                                                                                                                                     |                                                                                                                              |                                                                                                 |
|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Criterion                                            | Excellent (4)                                                                                                                                                                                                                                                                                                       | Proficient (3)                                                                                                               | Developing (1–2)                                                                                |
| Use of evidence from the unit                        | Uses specific data from the lessons accurately, such as the glacier area of about 1.35 km <sup>2</sup> in 1934 falling to less than 0.20 km <sup>2</sup> in 2023, the loss of over 90% of its mass since 1900, the Embu farmer's testimony, the Keeling Curve and the PhET observations, and links each to a claim. | Uses some evidence from the unit correctly but with limited detail or missing a number, and links most evidence to claims.   | Gives little or no evidence, or evidence is inaccurate or not linked to a claim.                |
| Physics of the greenhouse effect                     | Correctly traces shortwave radiation in, longwave infrared out, absorption and re-emission by greenhouse gases, and back-radiation, and clearly distinguishes the natural from the enhanced greenhouse effect.                                                                                                      | Describes the main steps of the mechanism with minor gaps or imprecise terms, and mentions the natural and enhanced effects. | Describes warming in general terms, with missing steps or confusion between types of radiation. |
| Causal chain and explanation of both phenomena       | Builds a complete connected chain from Kenyan human activity to greenhouse gas increase, to enhanced back-radiation, to warming, to glacier loss and altered rainfall, and explains why one cause accounts for both.                                                                                                | Builds a mostly complete chain with one step missing or weakly explained.                                                    | Gives isolated statements without a connected chain, or explains only one of the two phenomena. |
| Distinguishing ideas (ozone, correlation, mechanism) | Clearly separates ozone depletion from the greenhouse effect and the correlation shown in data from the explanation by physics.                                                                                                                                                                                     | Separates these ideas with some imprecision.                                                                                 | Confuses ozone depletion with the greenhouse effect or treats correlation as the explanation.   |
| Responses and communication                          | Suggests Kenyan responses placed at the correct step of the chain, uses the Montreal Protocol as evidence, and writes clearly with correct scientific terms and a relevant open question.                                                                                                                           | Suggests reasonable responses with limited link to the chain and mostly clear writing.                                       | Gives vague responses or unclear writing with few scientific terms.                             |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

**Production & Layout:** Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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